

Mathematics for New Technologies in Finance

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Mini-project

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This mini-project is due **Monday, May 11 at 23:59**, and can provide a 0.25 bonus on the final grade. While it is optional, we highly recommend you give it a go, as it summarizes a lot of the notions that have been studied so far. For correction and reproducibility, please use a random seed. If a result is clearly wrong but you do not have time to correct it, please state it and explain your reasoning. Finally, please be mindful of the length of your report - 3 pages is the maximum allowed (with plots).

The project setting and the results come from the broader framework treated in [1].

We consider the problem of maximizing an investor's asymptotic growth rate under model uncertainty, in the presence of stability, that is ergodic dynamics for the market. In particular, we study

$$\sup_{\theta \in \Theta} \inf_{\mathbb{P} \in \Pi} g(\theta; \mathbb{P}) \quad (1)$$

where Θ is the set of admissible trading strategies, Π is the set of admissible measures and $g(\theta; \mathbb{P})$ is the investor's asymptotic growth rate when they adopt the trading strategy θ and the dynamics of the market are governed by $\mathbb{P}$.

Let us fix an integer $d \geq 1$ and a non-empty set $E \subset \mathbb{R}^d$. The market consists of d risky assets with discounted prices $X = (X^1, \dots, X^d)$ taking values in E .

Consider the canonical space $(\Omega, \mathcal{F}, \mathbb{F})$, where Ω is the set of all E -valued continuous trajectories with the topology of locally uniform convergence, $\mathcal{F}$ is the Borel σ -algebra and $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ is the right-continuous filtration generated by X .

For simplicity here, let us assume that $(X_t)_{t \geq 0}$ is a d -dimensional Ornstein-Uhlenbeck process. As input, we give ourselves two symmetric positive definite matrices $c, \Sigma \in \mathbb{S}_{++}^d$ and a function $p : E \rightarrow (0, \infty)$ given by

$$p(x) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2} x^\top \Sigma^{-1} x\right) \quad (2)$$

that, you might recognize, is the density of an $\mathcal{N}(0, \Sigma)$ distribution. The matrix c serves as the instantaneous covariance of the process X .

Let us assume that X has ergodic density p . Define the wealth process at time $T > 0$ for

a strategy $\theta \in \Theta$ and with $\mathbb{P} \in \Pi$ governing the market dynamics as

$$V_T^\theta := \mathcal{E} \left(\int_0^T \theta_t^\top dX_t \right) \quad (3)$$

where $\mathcal{E}$ is the stochastic exponential.

The asymptotic growth rate of the portfolio under the measure $\mathbb{P}$ with strategy θ is defined as

$$g(\theta; \mathbb{P}) := \sup \left\{ \gamma \in \mathbb{R} : \liminf_{T \rightarrow \infty} \frac{1}{T} \log V_T^\theta \geq \gamma, \mathbb{P}\text{-a.s.} \right\} \quad (4)$$

You might here also recognize the log-utility problem presented in the lecture.

Let $A(x) = cp(x)$ be the averaged covariance matrix and consider the classes

$$\mathcal{D} = \left\{ \varphi \in C^2(E) : \int_E \left| \frac{\text{Tr}(A(x) \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} \right| dx < \infty \right\} \quad (5)$$

and

$$\Theta_0 = \left\{ \theta \in \Theta : \theta_t = \nabla \varphi(X_t), \varphi \in \mathcal{D} \right\} \quad (6)$$

Finally, define the growth rate invariance property: " $g(\theta, \mathbb{P})$ is independent of $\mathbb{P} \in \Pi$ for every $\theta \in \Theta_0$ ".

Part I.

1. What does Θ_0 represent?
2. For $\theta_0 = \nabla \varphi(X_t) \in \Theta_0$, express $\log V_T^{\theta_0}$ using Itô's formula and the ergodic property, and derive an expression for $g(\theta_0, \mathbb{P})$. Deduce that Θ_0 has the growth rate invariance property over Π . What does this mean for the problem we are trying to solve? How does it read now?
3. For $\varphi \in \mathcal{D}$ compactly supported, integrate by parts the following expression to make the Hessian vanish:

$$\int_E \frac{\text{Tr}(A(x) \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx \quad (7)$$

4. Recall what $\text{div} A$ is and write $A^{-1} \text{div} A$ as a function of p .
5. Since we considered p to be the density of a centered Gaussian, rewrite the optimization problem of 2. and show that it is Linear Quadratic (LQ). Solve it for φ .
6. What is the robust-optimal trading strategy? Justify.
7. Compute the robust-optimal growth rate. What do you observe?

Part II.

For the second part of the project, the goal is to robustify via adversarial training to compute the robust-optimal growth rate. Note that, in the general case, this is a non-trivial question. However, in the case of a multi-dimensional Ornstein-Uhlenbeck process that we consider here, it is more amenable.

Consider an Agent that trades in X and an Adversary, that we call "Nature", that decides the drift of X . For the optimization to be robust, it must be performed over all dynamics that induce the ergodic distribution p in the long run.

1. Write down the d -dimensional OU dynamics of X (in a general form). Which specific parameters within the drift term will the Adversary perturb to induce the worst-case scenario for the agent?
2. When modifying the drift, what constraint must Nature satisfy to maintain a valid stationary distribution? Specifically, how is the Fokker–Planck equation used to relate the drift matrix and the diffusion matrix to the stationary covariance matrix Σ ?
3. In terms of d , write down the number of degrees of freedom the Fokker–Planck equation allows. Justify your answer and argue.
4. Propose a design for the Agent and Adversary as neural networks and define the loss functions for this minimax game. Specify the architectures and justify why these choices are appropriate.
5. Formulate the training procedure to learn the robust-optimal strategy. How do you handle the potential instabilities of this objective?
6. Plot the learned strategy and report the computed robust-optimal growth rate. What do you observe?
7. Compare the analytical result of Part I. with the ML approach of Part II and comment on your findings.

References

- [1] David Itkin, Benedikt Koch, Martin Larsson, and Josef Teichmann. Ergodic robust maximization of asymptotic growth with stochastic factor processes. *Finance and Stochastics*, 2022. arXiv preprint arXiv:2211.15628.